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Human-Machine symbiosis in educational leadership in the era of artificial intelligence (AI): Where are we heading?

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Abstract

The issue of Artificial Intelligence (AI) and educational leadership has become a central topic following the shift to knowledge technology and digital literacy in recent years. Despite increasing attention to the topic among scholars and policymakers during the last decade, we lack a comprehensive review of AI and educational leadership. Therefore, by using bibliometric analysis of data derived from the Scopus database, this paper represents a systematic scoping of AI and educational leadership, tracing its development, characteristics, and knowledge accumulation within broad perspectives. Results identify key trends, geographic representation, topical foci, main concepts, themes, and their interconnections, including their implications for targeted Sustainable Development Goals. Implications, limitations, and future research reviewed in the studies are fully discussed.

Keywords

Artificial intelligence, technology, digital leadership, decision-making, administration, organization, future of work

Introduction

The history of Artificial Intelligence (AI) and educational leadership can be traced to the 1950s. Specifically, the term AI was coined in a scholarly event organized in 1955 in response to a question raised by Turing (1950) *can machines think?* (Huang et al., 2023). AI was conceptualized as machines and processes imitating human cognition and making decisions like humans (McCarthy et al., 1955). Although, Murphy offered a clear AI definition, which is the "applications of software algorithms and techniques that allow computers and machines to simulate human perception and decision- making processes to successfully complete tasks" (Murphy, 2019: 2). In other words, AI encompasses technology that can gather information and use that information to make decisions and complete tasks at a level similar to that of a human. Also, AI applications in education

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are growing exponentially (Chen et al., 2020). Some scholars continue to raise concerns that it will cause a dystopian future to come to fruition. To understand these concerns and how they differ from previous school-based applications of technology, a brief primer based on current AI technology and software starts with some of the following definitions: "AI refers to machine-based systems that can, given a set of human-defined objectives, make predictions, recommendations, or decisions that influence real or virtual environments" (UNICEF, 2021: 16). Interestingly, Holmes and Porayska-Pomsta (2023) divided the subject of AI in education into four segments. First, l*earning with AI* involves how AI can help students in the classroom, either through classroom applications or school administration systems. Second, *using AI to learn about learning* is a way that researchers and educators can use data mining and other AI-based tools to analyze educational data. Third, *learning about AI* is what you are doing by reading this book. It covers everything from teaching young children about data privacy to teaching graduate students the details of natural language processing. Finally, AI in education includes *preparing for AI* which is seen as the action steps people can take once they learn about AI.

Many fears around AI, on the other hand, are geared toward applications that could demonstrate super-human levels of intelligence and take actions dangerous to humans. This type of AI is characterized as strong or general intelligence AI, and while part of the AI research field, general intelligence AI is generally assumed to only be attainable in the distant future (Murphy, 2019).

Decision-making is also conceptualized as one of the four major theories in educational leadership (Griffiths, 1959). Educational leadership is "a general, abstract application of decision-making" (Tarter and Hoy, 1998: 212). It is an ability and process, guiding, coordinating and encouraging other people to approach the art of education and to implement quality teaching and learning, so that the activities carried out can be more effective and efficient in accomplishing educational goals. The significant shift towards knowledge technology and digital literacy in system delivery has imposed considerable strain on societal structures, encompassing the educational domain, as they grapple with current life conditions and ecosystem needs (Arar and Chen, 2021). This has led educational institutes to redesign or reimagine their delivery systems, leadership strategies, and pedagogical approaches in the current era, reflecting the new understanding of professional educational capital as a crucial concern (Arar et al., 2023a, 2023b). The issue of educational leadership in the AI space therefore has also become one of the central and current topics in educational leadership and management in recent years. More specifically, the role of educational leadership in school development is identified as critical (Aldridge and McLure, 2023), especially in improving the quality of thinking, learning, and management (Arar et al., 2023b). Subsequently, AI technologies have been rapidly adopted by educational leaders leading to a reshaping of the field of education leadership. It is therefore important to investigate human-AI symbiosis in educational leadership.

Human-AI symbiosis in educational leadership

Human leaders make decisions using their senses (i.e., sight, hearing, touch, smell and taste) to collect incoming information, which is then processed by the brains' different systems, including the attentional system, memory system, motivational system, emotional system, and cognitive system. AI also uses the same approach where the term "artificial" resembles the way humans make decisions through their biological brains (Ullman, 2019). Specifically, AI uses different sensors (cameras, microphone, etc.) to collect data and artificial neural networks (a simulation of how neurons are organized in the brain) to analyze the data and make decisions.

With this similarity, yet differences, human educational leaders and AI can collaboratively work together to enhance decision-making and management, thus increasing the effectiveness of educational institutions. Specifically, AI can make use of its huge computational power to rapidly analyze big data and to draw conclusions. In fact, a huge amount of data is collected by educational institutions (e.g., exam scores for each subject, absence, demographic, etc.), where educational leaders often experience difficulties in analyzing them, especially since this data is not necessarily expressed in linear relationships but are "multi-level, processual, contextual and interactive" (Marion and Uhl-Bien, 2001: 631). Unlike AI, human cognition is limited by biological needs and can be affected by several factors like hunger, fatigue, stress, etc. Therefore, AI can be used as a data-driven approach to support educational leaders. Human educational leaders, on the other hand, are better than AI at making value based moral decisions. Feelings like empathy, guilt, regret, etc., are crucial in making decisions (Greene, 2013). While AI might be used to detect humans' feelings it still cannot make decisions based on feelings. Hodgkinson (1978) further coined decision making as "moral leadership" which is an "administrative logic" of a new order. For instance, school leaders pointed out that it is morally wrong to have uniform expectations for student achievement (Frick, 2009). Therefore, when making decisions, humans usually ask "what is the right thing to do?" while AI asks, "what is the next step to be done based on the data analysis?".

Based on the unique features of humans and AI, collaborative intelligence, i.e., the combination of human intelligence and artificial intelligence, can be established to increase educational leader-AI symbiosis, thereby enhancing decision-making and reshaping the future of educational leadership. For instance, Denden et al. (2019) developed an intelligent system which predicts "at-risk" students, those who might fail to pass their final exams, based on their online learning data. The system then informs the teacher about those predictions so that she can review them and make the appropriate educational interventions accordingly. Additionally, Intelligent Tutoring Systems (ITSs) are now being used in collaboration with human tutors to enhance learning experiences and outcomes (Wang et al., 2023).

However, we propose that the challenge lies in the notion of considering humans and technology as separate entities. The integration of knowledge technologies into educational enterprises often occurs without sufficient consideration for the learner (UNESCO, 2015). When educational policies and practices are formulated without addressing the learner's constraints, the result is a continuous failure. The three Frontiers of Learning—Curriculum, Pedagogy, and Time-Space of Learning—constitute challenges in reimagining the educational leadership role in the AI environment. In this context, educational leaders should be recognized as key players in the process of guiding schools in the AI space and conditions (Arar and Chen, 2021).

Conceptual framework

Artificial Intelligence (AI) is defined in various ways, each highlighting different aspects of its functionality and impact. The Organization for Economic Cooperation and Development (2024) describes AI as machine-based systems that generate outputs—predictions, content, recommendations, or decisions—based on input to influence environments. This technical definition emphasizes AI's functional capabilities and varying levels of autonomy and adaptiveness.

UNICEF (2021) contextualizes AI within human environments, focusing on machine-based systems that, given human-defined objectives, interact with and influence real or virtual environments. These systems often operate autonomously and adapt by learning from their context. In addition, Holmes and Porayska-Pomsta (2023) define AI as a complex sociotechnical artefact constructed through social processes. This perspective highlights the social construction of AI and the importance of how intelligence is defined and enacted.

Intelligence itself has evolved in definition, particularly in education. Gardner's theory of multiple intelligences suggests that intelligence involves solving problems or doing something valued in one or more cultures (Morgan, 2021). This broad view implies that even common technologies like smartphones might have been considered intelligent a century ago.

These overlapping definitions contribute to the current educational experiences of using intelligent technology. AI applications that solve problems or make predictions have long been integrated into classrooms. However, specialized AI applications today have a broader reach and impact, affecting various stakeholders differently. Understanding these changes requires starting with the students' experience. In the same vein, Holmes and Porayska-Pomsta (2023) categorize AI in education into four segments: learning with AI, using AI to learn about learning, learning about AI, and preparing for AI. Learning with AI involves classroom and administrative applications. Using AI to learn about learning utilizes data mining and AI tools for educational data analysis. Learning about AI covers educating students about AI, from data privacy to natural language processing. Preparing for AI involves actionable steps after learning about AI.

Integrating these perspectives, the conceptual framework for AI in educational leadership encompasses functional, contextual, and sociotechnical definitions of AI, and the evolving nature of intelligence. Educational leaders must leverage AI to enhance learning experiences, improve administrative efficiency, and inform teaching and research. They also need to educate students about AI and prepare them for a future dominated by intelligent technologies (Papa and Jackson, 2022).

In summary, understanding AI in educational leadership requires recognizing its technical capabilities, contextual interactions, and sociotechnical construction. This framework supports educational leaders in using AI to improve learning outcomes, administrative functions, and student readiness for a future with pervasive AI.

Research gap and study objectives

Despite extensive literature discussing AI and education across diverse fields, there is currently no widespread consensus on the meaning of leading education in the AI era (Arar and Chen, 2021). The pivotal question arises: How is it possible that these technological advancements have left learning institutions virtually unchanged? Can knowledge technologies genuinely transform learning institutions into effective, fair, and functioning complexes? Milton and Al-Busaidi (2023) urged educational leaders to be more prepared, adaptable, and updated to deal with AI as it reshapes educational leadership. Ghamrawi et al. (2024) concluded that the impact of AI on educational leadership depends on how it is implemented and integrated into an education system.

However, despite the opportunities that AI might create to reshape educational leadership, scant information exists in this regard (Wang, 2021a, 2021b). Moreover, Elo and Uljens (2023) assert that educational leadership research could lead to a variety of academic trends and approaches, especially in the context of technology. Peifer (2022) mentioned the rapid development of AI requires further research on its impact on leadership to support institutions with practice-proven guidelines and solutions.

Besides, considering that AI in educational leadership mostly aims to improve educational experiences, augment learning outcomes, and maximize inclusivity, which fall under the sustainable development goal "quality education", no other studies have analyzed research on AI in educational leadership by approaching the topic in question from the perspective of Sustainable Development Goals (SDGs).

The scarcity of bibliometric reviews analyzing the impact of AI in educational leadership makes it difficult to determine whether the existing knowledge base can inform the dramatic changes in this landscape. In line with the broader picture of research production and bibliometric analysis in educational leadership and management (Gümüş et al., 2021; Hallinger & Kovačević, 2019, 2021), this is a significant lacuna that needs to be addressed. In line with Donthu et al. (2021) argument that bibliometric analysis offers a powerful and rigorous method for exploring and analyzing large volumes of scientific data, making it particularly valuable for understanding the rapidly evolving fields of AI and educational leadership.

It is against this backdrop that the current bibliometric analysis was conducted. The use of bibliometric analysis in understanding AI and educational leadership is supported by its ability to comprehensively map existing literature, identify research gaps and emerging trends, compare the impact of different research areas, and provide methodological rigor. These advantages make bibliometric analysis an essential tool for researchers and practitioners aiming to stay at the cutting edge of these dynamic fields.

Given that AI has rapidly evolved from expert systems to machine learning, deep learning, and now generative AI, it is crucial to investigate these trends and approaches. To fill this research gap, this study conducts a bibliometric analysis of research on AI in educational leadership to systematically reveal future research trends and topics that various stakeholders should consider to facilitate the adoption and implementation of AI in educational leadership. To our knowledge, no study has systematically reviewed the literature to provide a roadmap for how AI might reshape educational leadership and identify future research directions. Specifically, this study addresses the following main research questions:

RQ1. What is the volume and geographic distribution of research on AI in educational leadership in terms of publication year, country, collaboration, citation, and publication venues?

RQ2. What is the discourse on AI in educational leadership in terms of keyword frequency trends, and themes?

RQ3. What is the conceptual structure stemming from the thematic mapping (the topical foci and frequently used concepts) of the knowledge base on educational leadership and AI?

RQ4. What is the distribution of research on AI in educational leadership in terms of the targeted sustainable development goals?

Method

Developments in data mining and social network analysis, alongside advancements in related tools, now enable comprehensive reviews of the knowledge base in specific fields or topics based on extensive research studies (Gümüş et al., 2021; van Eck and Waltman, 2017). Donthu et al. (2021) argue that bibliometric analysis is a popular and rigorous method for exploring and analyzing large volumes of scientific data, particularly valuable for unpacking the evolutionary nuances of a field and identifying emerging areas. This methodology has gained significant traction in business research (e.g., Donthu et al., 2021; Khan et al., 2021) and in educational leadership and management, given the availability and accessibility of bibliometric software and databases like Scopus and Web of Science.

One key argument for using bibliometric analysis in this context is its ability to provide a comprehensive overview of the existing literature. By systematically mapping out the body of research, bibliometric analysis can identify the most influential studies, authors, and journals in the field. This helps to establish a foundational understanding of the current state of knowledge, pinpointing seminal works and key contributors (Donthu et al., 2021), especially as this method relies on massive, objective datasets (e.g., citations, publications, keyword occurrences) and combines objective performance analysis with subjective thematic analysis.

Notwithstanding its merits, Ellegaard and Wallin (2015) emphasize that bibliometric analysis excels in deciphering and mapping cumulative scientific knowledge and the evolutionary nuances of well-established fields, making sense of large volumes of unstructured data rigorously. Well-conducted bibliometric studies provide a solid foundation for advancing a field in novel and meaningful ways, enabling scholars to gain a comprehensive overview, identify knowledge gaps, derive novel ideas for investigation, and position their contributions effectively within the field.

In recent years, the use of bibliometric methods has increased in systematic review studies, including those reviewing scholarship on educational leadership (Gümüş et al., 2021; Hallinger et al., 2020; Wagner, 2024). This study uses bibliometric analysis to unveil trends in educational leadership in the era of AI. This is particularly important for AI in educational leadership, where staying ahead of new developments can provide a competitive edge and ensure that educational practices are at the forefront of technological advancements. Moreover, bibliometric analysis offers a method to compare the relative impact of different research topics and methodologies. This comparative analysis helps to highlight which areas of AI and educational leadership are producing the most significant findings and which methodologies are proving most effective. For policymakers and educational leaders, this information is crucial for making informed decisions about where to allocate resources and focus efforts. Bibliometrics has evolved beyond frequency and citation analysis to include advanced methods such as network analysis, machine learning, and text mining, ensuring data triangulation (Thurmond, 2001) and providing deep insights into a given topic (Schöbel et al., 2021). Various studies have conducted bibliometric analyses to reveal trends in emerging topics such as remote special education during COVID-19, open educational resources, and learning analytics (Arar and Chen, 2021; Tlili et al., 2021a; Tlili et al., 2021b).

Identification of sources

To ensure selecting the most relevant literature to be reviewed, this study followed the Preferred Reporting Items for Systematic Reviews and Meta-Analysis (PRISMA) guidelines (Page et al., 2021). The data for this study was collected from Web of Science (WoS), as it is a popular database for bibliometric research on leadership (Gamarra and Girotto, 2022; Vogel et al., 2021). While several studies highlighted that covering journal papers only would lead to quality results, other studies also pointed out that excluding other types of documents, such as conference papers and books, might hinder the full understanding of research on leadership (Hsieh et al., 2023; Gümüş et al., 2021). To reflect a comprehensive view and understanding of educational leadership research in the AI era, this present study, therefore, covered all types of documents without any specific time span. Specifically, the following search query was conducted in Web of Science Core Collection, specifically in the “All field” search.

(Artificial Intelligence) AND ((educational leadership OR leadership in education OR leadership in schools OR leadership in universities OR leadership in institutions)).

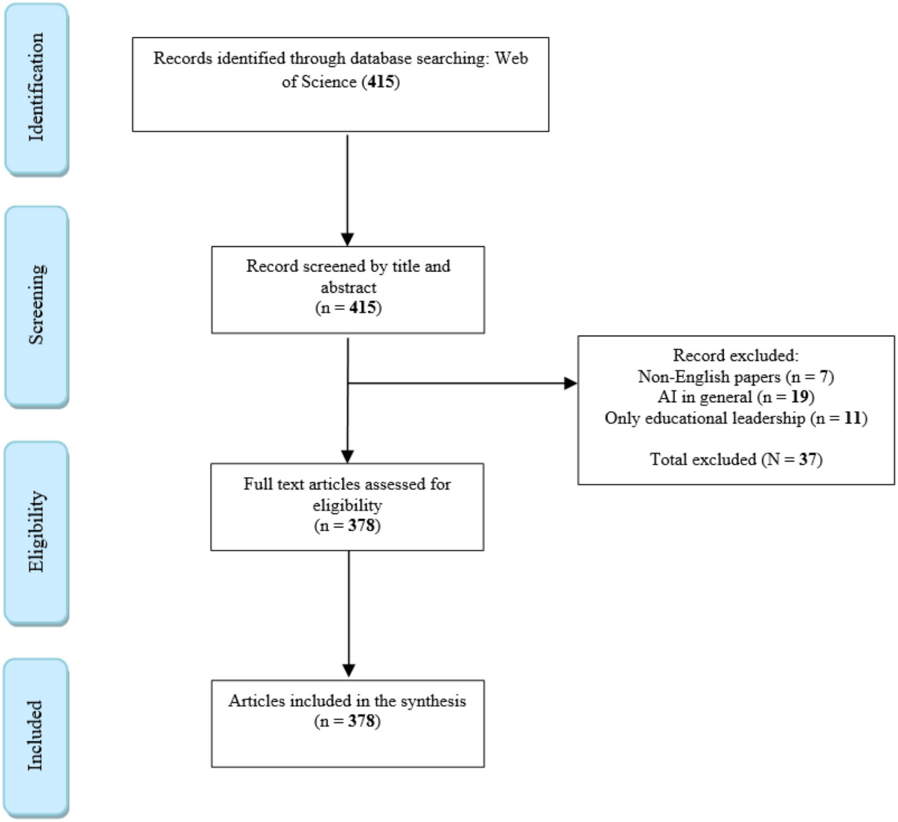

Figure 1. PRISMA flowchart for the search protocol.

A study was excluded if: (1) it discusses AI in education without a focus on educational leadership; (2) it discusses leadership but not in education (e.g., in work); and (3) a study is not in English. As a result, 378 studies were obtained and included in this present study. Figure 1 presents the PRISMA flowchart (Page et al., 2021) of the study selection process.

Data processing and analysis

The analysis conducted in the present study goes beyond the descriptive results (major contributors, highly cited papers, and organizations) to also conduct various techniques (clustering, science mapping, etc.) that could provide more in-depth results and analysis (Donthu et al., 2021; Mukherjee et al., 2022). In line with Mukherjee et al. (2022), the authors of the present study believe that figures and diagrams can also support advancing theory if they serve as building blocks for theory advancement. Accordingly, word co-occurrence was conducted, which aims to analyze relationships between words or phrases, hence revealing more insights on a given topic focus (Van den Besselaar and Heimeriks, 2006). Additionally, a thematic map was applied (Tlili et al., 2024) to reveal trending topics related to AI and educational leadership.

The analysis of data was performed with *R! Programming language* using the *Bibliometrix* library. Bibliometrix is an open-source tool for quantitative research in scientometrics and bibliometrics that includes all the main bibliometric methods of analysis (Aria and Cuccurullo, 2017). In addition, the built-in web page analysis interface of Web of Science database was also used to further complement the obtained results from *R!*.

All 378 articles were first downloaded as a bib file (.bib) from the WoS and then uploaded to the *R!* software. The data was then imported and processed by the software to analyze the results and visualize them as reported in the results section (see Section 3).

Results and discussions

The results are presented and discussed according to each of the research questions.

Distribution in terms of publication year and type, country, citation, and publication venues

Figure 2 presents the distribution of AI and educational leadership research according to the publication year. It is seen that the first research was conducted in 1986 with an annual growth per year of 12.42%. The first study (Halff, 1986), published in *Educational leadership: journal of the Department of Supervision and Curriculum Development*, discussed how AI instructional applications will revolutionize educational institutions and the way they operate. Particularly, the researcher mentioned several AI technologies, such as Intelligent Tutoring Systems (ITS) and expert systems, will have profound impact in leading teaching practices to a new level. Not so long after it, another study (Nelson, 1989) appeared discussing how AI will change knowledge acquisition through knowledge engineering. This catalyzed research on AI and educational leadership, where Standley (1999) conducted the first workshop on building technology leadership skills

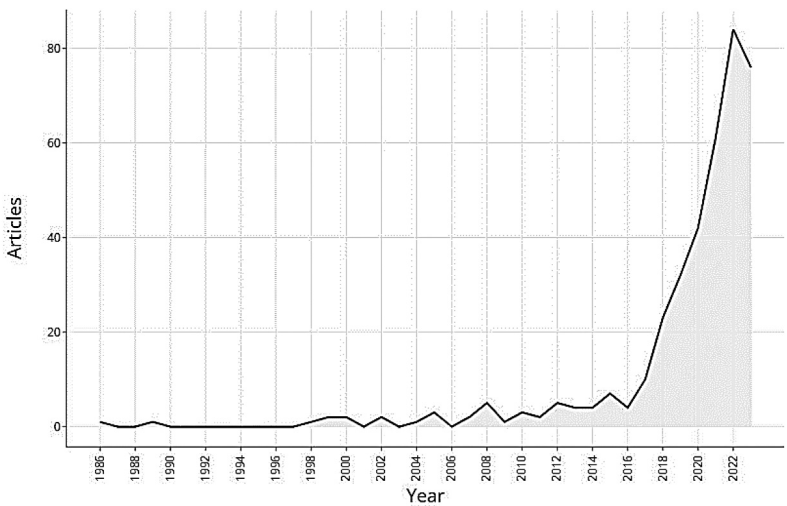

Figure 2. Distribution of research on AI and educational leadership by year.

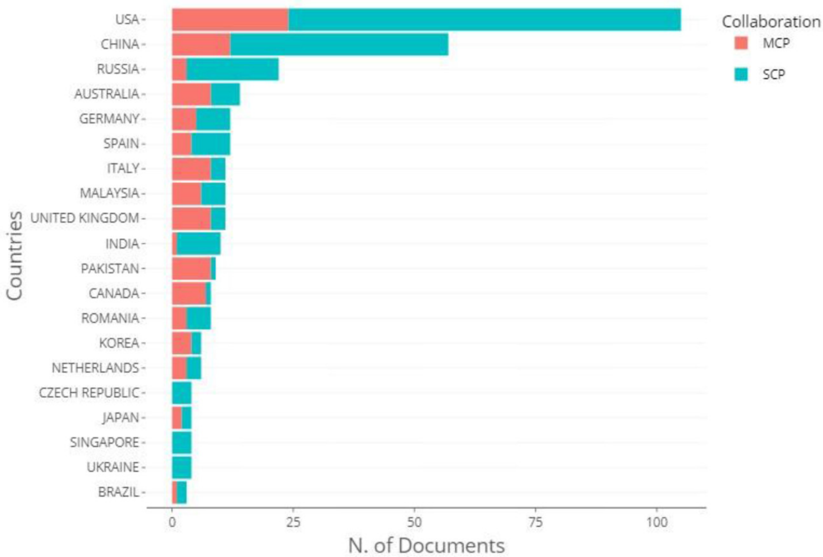

Figure 3. Distribution of research on AI and educational leadership by country.

in the 7th International Conference on Computers in Education (ICCE 99). Starting in 2016, a sharp peak is seen in the research publication on AI and educational leadership. This could be attributed to the maturity of several AI-based technologies, such as machine learning, big data, and learning analytics, which took AI technology from modes of linear development (e.g., Moore's law; Moore, 1965) to conducting more complex managerial tasks. This catalyzed the integration of AI in educational leadership for different objectives, including: (1) providing suggestions and guidance in corporate strategy development (Herse, 2021), as well as in research and development (R&D) fields (Cockburn et al., 2018); (2) providing assistance in managing employees (Gerpott et al., 2018); and (2) predicting leaders based on group communication (e.g., Zhang and Zhou, 2017). The AI implications on educational leadership have also been discussed from several stakeholders' perspectives, such as school managers (Dagli et al., 2023) and teachers (Ghamrawi et al., 2024).

Φιγυρε 3 presents the distribution of AI and educational leadership research according to the corresponding author's country. To aid in visual presentation of the data, only the top 20 countries are presented in Figure 3. It shows the United States had the highest number of publications (105 studies), followed by China (57 studies), Russia (22 studies) and Australia (14 studies). This could be explained by the fact that these countries have already developed a strong adoption and implementation of AI from educational perspectives, and this catalyzed their adoption and implementation of AI in educational leadership. In addition, the data reveals the United States also has the highest multiple country publications (MCP) in which authors belong to different countries, and such publications represent inter-country collaboration, i.e., international collaboration. Single country publications (SCP), on the other hand, imply that all authors belong to the same country and such publications represent intra-country collaboration. Interestingly, there is an absence of countries from the African region or the Middle East (see Figure 4), and this might be attributed to the low adoption level of AI in these areas (Eke et al., 2023).

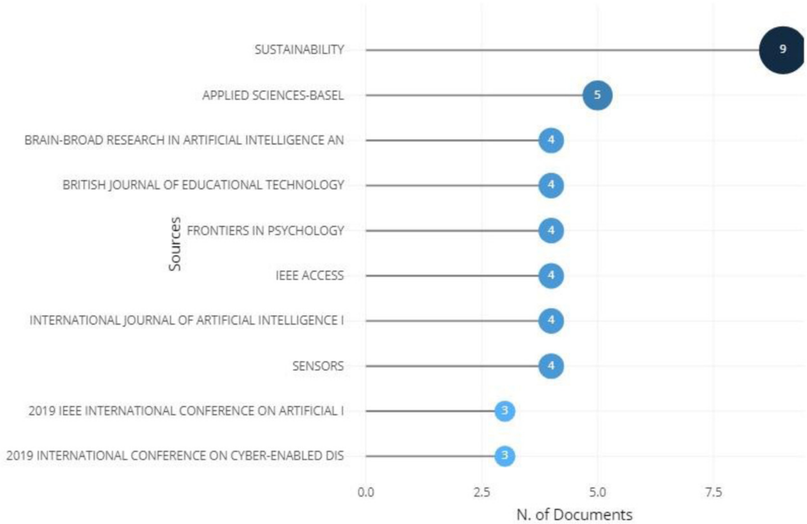

Figure 4. Top 10 venues for AI and educational leadership.

Figure 4 lists the top 10 venues where researchers have published their AI and educational leadership research. This analysis is important since it informs stakeholders about the leading journals in the field and where they can find the relevant literature, or where they can aim to publish their research. It is seen that the top journal on the list, *Sustainability*, is cross-disciplinary. This implies that adopting AI in educational leadership requires competencies from different fields, including computer science, security, education, educational leadership, among others. In this context, the European Commission (2019) stated that interdisciplinary approaches can help individuals develop a holistic understanding of AI technologies and their potential implications. Additionally, it is seen that most of the venues are computer science journals with an AI focus (as seen in the journal title).

Distribution in terms of keyword frequency and trends, and themes

Figure 5 presents the top 30 authors' keywords captured in the word cloud. The size of the word indicates its frequency of appearing in the reviewed studies. In addition to leadership and education, it is seen that machine learning has been frequently used. This finding aligns with other research showing machine learning is a subset of AI which has been used with data-driven approaches to assist school leaders in making decisions (Wang, 2021a). Coyle and Weller (2020) further argue that machine learning might lead to policy reform for both its positive and negative perspectives. AI-powered technologies, such as blockchain and robotics (see Figure 5) have recently started being used in educational leadership. For instance, since leadership is built on trust among people, blockchain is introduced, through its distributed and secured ledger, to increase trust

Figure 5. Top 30 word cloud based on author keywords.

between people, thus enhancing leadership decisions in terms of transparency; for instance, when choosing with whom to work/collaborate as people's behavior on the blockchain system are traced and visible (Chartier-Rueg and Zweifel, 2017; Lipshaw, 2020). Interestingly, generative AI, specifically ChatGPT (see Figure 5), has also appeared in the discussion of how AI can reshape educational leadership. Fullan et al. (2023) indicated that the rise of generative AI will redefine the nature of education delivery in schools. Generative AI can support school leaders in routine administrative tasks and enable them to focus on advanced pertinent issues (Harris and Jones, 2023).

The keywords depicted in Figure 5 also show that AI in educational leadership has been used for different purposes, including monitoring, innovation and creativity, modeling, and communication. AI can facilitate educational leaders' work, enable them to solve problems, and empower them to be creative in the process of problem solving (European Commission, 2022). Yang et al. (2022) encouraged school leaders to innovate and create stronger collaboration with AI experts to improve their professional development and daily practices using AI-based tools. This also will aid human judgement and identifying algorithmic bias (Yang et al., 2022).

To achieve a deeper understanding of how these keywords appeared and discover interpretable relationships and research topics related to the research on AI and educational leadership, a network of words based on the co-occurrence of keywords was created (see Figure 6). The nodes represent keywords and were labeled accordingly, and the edge between the two nodes represents the co-occurrence between those keywords. The size of a node and label indicates the frequency of a keyword in the dataset, whereas the thickness of an edge indicates the co-occurrence frequency between keywords. A greater thickness demonstrates that the keywords are closely related to each other. The color of the node shows the cluster with which the keyword is associated. Each cluster belongs to a research theme represented by the keywords and their links. Figure 6 shows that six different clusters in different colors have emerged.

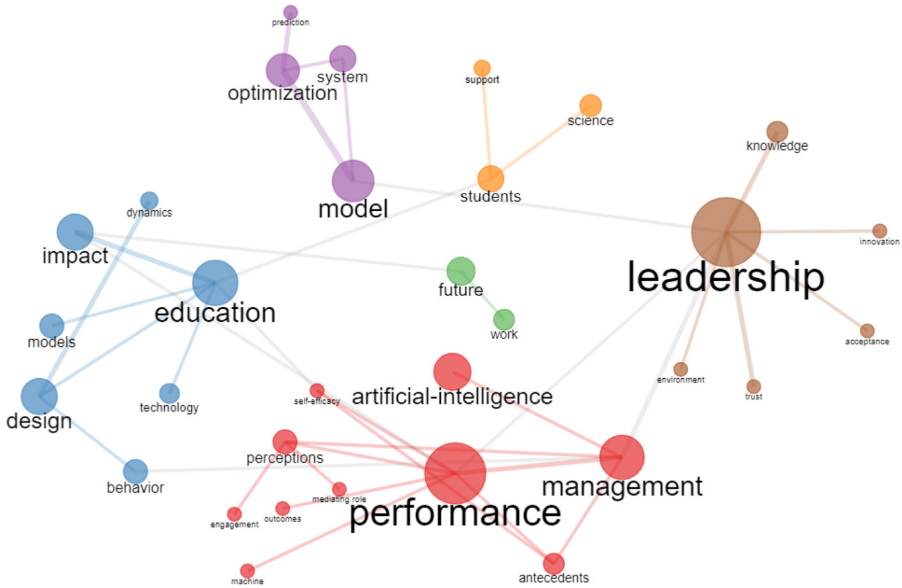

Figure 6. Co-occurrence network author keywords.

The first brown cluster focuses on the leadership concepts that AI might improve (See connected nodes in Figure 6: *trust*, *knowledge*, *innovation*, *environment*, *acceptance*) for better decision making and management, hence better performance. One of these concepts is trust which has been reported crucial in leadership (Kouzes and Posner, 2007). As Currall and Epstein (2003: 203) stated "if properly developed, trust can propel [organizations] to greatness. Improperly used, trust can plant the seeds of collapse." Therefore, one of the ways where AI is used to improve leadership trust is analyzing educational data to increase efficiency and accuracy in educational leaders' decisions. This approach is also known as data-informed decision making (DIDM) (Wang, 2021b). DIDM can also foster leadership innovation as the AI results might reveal hidden and effective decisions that might not be seen with naked eye. However, having such fair and accurate decisions, based on DIDM, requires quality big data and responsible algorithms.

The second purple cluster focuses on the models that AI could create (See connected nodes in Figure 6: *optimization, system, prediction*). Regarding optimization, AI is used to classify leadership methods and types. These classifications may be used to optimize current leadership styles (Walczak, 2016). Moreover, the integration of AI in leadership can create optimized teams by analyzing teams and potential members to create the best combination of characteristics and expertise possible. Leaders could face different challenges when they integrate AI systems in their practices. Some of these challenges are presented in systems bias and distrust in data gathering and algorithms (Okunlaya et al., 2022). Therefore, Soni (2020) advised leaders to properly and adequately implement AI systems into their institutions, to avoid conflict situations. AI can improve prediction, which is vital for leaders to make decisions and plan for the future (Goldfarb and Lindsay, 2022; Shafiabady et al., 2023). The accurate predictions resulting from the use of AI models will support leaders to progress and improve the organizational performance and creativity (Mikalef et al., 2023; Li et al., 2022).

The third green cluster focuses on future of work in the era of AI and leadership (See connected nodes in Figure 6: *future, work*). In the next ten years, it is expected that AI will automate about 40% of the noninstructional tasks performed by teachers, such as tracking student progress (Herold, 2019). AI will also reshape the instruments of instructional tasks such as classroom management tools and tutors that supported personalized instruction and adaptive learning and testing (Sparks, 2017). In a survey conducted by Bushweller (2020), teachers believed that AI could support them in implementing repetitive, time-consuming tasks, such as taking attendance and making copies. Bryant et al. (2020) reported that AI could free up about one third of teachers' working hours, allowing time for especially human driven tasks such as focusing on the social interactions and developing the emotional relationships necessary to inspire students and collaborate with other teachers. These tasks are challenging for school leaders as they have the potential of increasing teacher burnout.

The fourth red cluster focuses on perceptions of AI and leadership (See connected nodes in Figure 6: *mediating role, outcomes, engagement*). Leaders' perceptions regarding implementing AI in their institutions are still unknown due to the lack of conceptual awareness about AI (Neher et al., 2023), especially with regarding the mediating role of AI as it facilitates basic workplace transformation in the face of challenges (Chryssolouris et al., 2023). The introduction of AI affects leaders and employees' outcomes as AI could be implemented rapidly and effectively by utilizing already-existing software tools or algorithms. Moreover, AI creates research strategies for leaders to achieve the desired outcomes (Chen et al., 2023; Raisch and Krakowski, 2021). Leaders who integrate AI in the employees' professional growth enhance employee engagement and build high-performing teams (Rožman et al., 2023).

Conceptual structure

To gain more insights about the future research topic of AI and educational leadership, thematic mapping was conducted. As depicted in Figure 7, the major themes are divided into four different quadrants in a two-dimensional graph, namely the x-axis (representing density) and y-axis (representing centrality). Basic themes (placed in the lower-right quadrant, see Figure 7) indicate that they are significant for the research on AI and educational leadership, but less developed. They cover: *Artificial Intelligence management impact*, *care risk intervention*, *internet security*, and *design behavior dynamics*. Motor themes (placed in the upper-right quadrant, see Figure 7) present well-developed and significantly structured research topics on AI and educational leadership. They include *trust perception*, *leadership performance*, *selection discovery* and *big data*. Emerging or disappearing themes (placed in the lower-left quadrant, see Figure 7) indicate that they are at the infancy stage of development with marginal importance, majorly demonstrating either promising or disappearing themes. They include *attitude*, *adoption*, *classification*, *age*, and *language*. Finally, niche themes (placed in the upper-left quadrant, see Figure 7) indicate strongly developed internal ties (high density) with unimportant external links and so are of only limited importance (low centrality) for the research field on AI in educational leadership. They include *bias*, *search algorithm*, *research productivity*, and *cognitive impairment*.

Distribution of studies based on the targeted sustainable development goals

Based on the Web of Science built-in analysis tool, Table 1 presents the distribution of research on AI in educational leadership according to the targeted Sustainable Development Goals (SDGs)

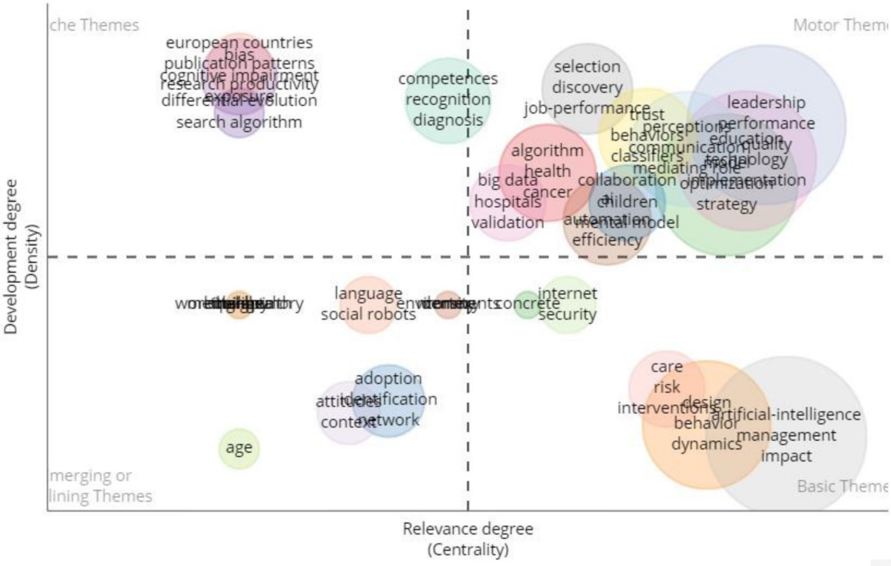

Figure 7. Thematic map analysis of research on AI in educational leadership.

Table 1. Distribution of research on AI in educational leadership according to targeted SDGs.

Sustainable Development Goals (SDG)	Record Count	% of 378
Quality Education (SDG 4)	72	19.048%
Good Health and Well Being (SDG 3)	66	17.460%
Industry Innovation and Infrastructure (SDG 9)	30	7.937%
Sustainable Cities and Communities (SDG 11)	27	7.143%
No Poverty (SDG 1)	11	2.910%
Responsible Consumption and Production (SDG 12)	11	2.910%
Climate Action (SDG 13)	11	2.910%
Zero Hunger (SDG 2)	5	1.323%
Decent Work and Economic Growth (SDG 8)	5	1.323%
Reduced Inequality (SDG 10)	5	1.323%
Life on Land (SDG 15)	4	1.058%
Gender Equality (SDG 5)	3	0.794%
Affordable and Clean Energy (SDG 7)	3	0.794%
Clean Water and Sanitation (SDG 6)	2	0.529%
Life Below Water (SDG 14)	1	0.265%

(UNESCO, 2019). It should be noted that 41% of the studies did not contain data in the field being analyzed.

Table 1 reveals that SDG 4 (quality education) was the most targeted SDG (19%) when integrating AI in educational leadership. This is understandable as most educational leaders have tried to integrate AI to enhance education and its outcomes which is in line with the SDG 4 goals. In this context, The United Nations policy on AI as a tool points out that AI can help achieve the SDG 4 particularly because it provides new possibilities for educators, enabling them to utilize technology for personalized learning, making education more effective and inclusive.

The second targeted SDG 3 (Good Health and Well Being) comprises 17.5% of the total record. This focus can be explained by the fact that the health field is a sensitive field where timely and accurate interventions are crucial, hence integrating AI to support educational leaders is much needed. For instance, AI algorithms can analyze enormous amounts of data to simulate the best procedures to treat patients based on their symptoms, medical history, family health records, among others (Bohr and Memarzadeh, 2020). In health and medical education, the effectiveness of AI applications is explored in training, learning, simulation, curriculum, and developing new assessment tools; the future physician will likely need to incorporate AI in certain patient care tasks, like history-gathering, visual diagnosis, information synthesis, patient outcomes, reducing costs, and decision-making (Azer and Guerrero, 2023; Gracias et al., 2023). That is why Makridis and Mishra (2020) pointed out that there is a significant positive association of AI and well-being.

Almost 8% of the studies discussing AI and educational leadership research focused on SDG 9 (Industry Innovation and Infrastructure). This is because, as pointed out in our aforementioned results, educational leaders have integrated AI to also foster innovation and creativity. In this context, Mhlanga (2021) discovered that AI can develop the financial sector by enabling all individuals to participate in the mainstream economy. Gonzales (2023) found a positive relationship between AI and economic growth.

Of course, achieving all of the above SDGs, as well as other SDGs, can lead to developing Sustainable Cities and Communities (SDG 11). However, Table 1 reported that few studies focused on SDG 11, which points to the need for more research in this regard. Particularly, it is important that future research studies go beyond the micro and meso levels to also investigate the macro level in terms of the AI impact. For instance, topics to consider are how educational systems and institutions might safely adopt and exploit machines, and how society could take advantage of AI for the common good (Huang et al., 2023).

Conclusions, future directions, and implications

This study conducted a bibliometric analysis to investigate the use of AI in educational leadership. The results obtained revealed that AI can support educational leaders and create massive opportunities that can enhance future education and outcomes, thus revolutionizing the education field. However, it is crucial to approach the adoption of AI with caution, as this study reveals several limitations that could potentially lead to actions that value co-destruction rather than co-creation. The findings of this study prompt consideration of the implications and challenges associated with integrating AI into educational practices. In light of the obtained findings, various avenues for future research are suggested, which will be explored in the subsequent sections to provide a comprehensive understanding of the subject.

Embrace digital transformation within educational institutions

The obtained results revealed that the adoption of AI in educational leadership requires that educational institutions should first embrace digital transformation. This is because, as our results revealed, data is considered as the fuel for AI-based systems to make decisions accordingly. The idea that knowledge can flow to the learner anywhere at any time is central to so-called distance learning. This technology threatens the very existence of formal learning institutions based on the need for learners to follow the physical location of knowledge and its formal agents. The scenario that suggests that schools and academia will not become obsolete assumes that learners can acquire knowledge without the organized time and space provided at school. Therefore, educational institutions must digitize their tasks (e.g., change from analog or physical to digital forms) and data (rely on digital data and traces instead of paper-based data) to harness the power of AI in educational leadership. Embracing digital transformation in educational institutions requires a comprehensive and detailed plan that involves various stakeholders (e.g., school leaders, teachers, administrators, students, etc.) to cater to the needs of everyone during this transformation.

Raising awareness and developing regulatory frameworks and policies

The findings revealed that several regions are still lacking behind in terms of adopting AI in educational leadership. Therefore, it is important that more governmental policies should be developed to raise awareness about the opportunities that AI might bring to the field of educational leadership. Particularly, educational leaders could disseminate a culture of AIED among teachers, students, and administrators. This kind of culture may include the definition, principles, ethics, and best practices of AI in education. On the other hand, more ethical and regulatory frameworks should be developed to protect users, for instance, in terms of privacy and also to define the working regulations, including legal responsibilities if an educational institution wants to include AI in its decision-making system. For instance, who could be sued if a mistake is made when using AI in educational institutions; the educational leader (e.g., the dean) or the developers of the AI system. More research is therefore needed in this line of research.

Developing responsible and human-centered ai

The findings revealed that the integration of AI in educational leadership requires considering equity, transparency, and accountability. In addition, educational leaders should work to identify and eliminate biases in data and algorithms. Buolamwini and Gebru (2018) further noted that AI systems should be trained on diverse and representative datasets to avoid bias and discrimination. It is, therefore, crucial to develop responsible AI in educational leadership. Responsible AI is concerned with the design, implementation, and use of ethical, transparent, and accountable AI technology to reduce biases, promote fairness and equality, and help facilitate the interpretability and explainability of outcomes.

Effective human-machine collaboration

The findings of this study revealed that AI may take over redundant tasks in educational leadership, raising concerns about an increase in the unemployment rate. Other studies, on the other hand, highlighted the need for the human touch in making decisions regardless of the AI advancement. It is, therefore, important to investigate how humans and machines (i.e., AI tools and systems) can collaborate together to enhance decision making. Kaber (2018), in this context, mentioned that one of the core questions in human-machine collaboration is "Who does what?". In a study investigating the effects of humans collaborating with machines, Nass et al. (1996) revealed that the "effects of being in a team with a computer are the same as the effects of being in a team with another human." In the same vein, de Vreede and Briggs (2019: 103) stated that, in the future, "artificial agents will become fully functional members of teams." This implies that AI should not replace humans and take over every role within a team when making decisions. Rather, they should be designed to fulfill certain activities which are most fitting and effective, and, thus, complement or augment the advantages of humans (Dellermann et al., 2019). Building on this perspective, it is crucial to investigate how human leaders can team up with AI to enhance decision making and achieve better learning outcomes.

Upskilling

This present study pointed out that educational stakeholders need upskilling in order to make full use of AI in decision making. This can be accomplished through acquiring strong knowledge of AI's concepts and applications in education via training programs designed for multiple roles and tasks. Hands-on experiences and cooperation with AI experts enable the learning of educational leaders. Emphasizing ethical considerations and responsible AI use is crucial. The adaptability to evolving AI trends is a vital component of ensuring significant integration and improvement in educational settings. For instance, ChatGPT does not require many technical or Information and Communication Technology (ICT) competencies, however, it requires more critical thinking and question-asking competencies to get the best results (Salha and Affouneh, 2023). Some research makes the argument that "AI as a historical future promises 'a disconnection' from past and present, and it cites that departure as the source of the possibility that even the most intractable political problems can be solved not by carefully unpacking them but by eliminating all of their priors" (Hong, 2022: 374). We suggest that transformative change requires recognizing outdated assumptions, updating teacher training, and clearly defining problems to which technologies can be applied.

Limitations

This study has several limitations that should be acknowledged and further researched. Firstly, although it provides significant information regarding knowledge structures, the substantive findings are derived mainly from the reviewed documents, and the mapping is constructed based on the 'metadata of selected and limited documents' (Gümüş et al., 2021: 95). Secondly, the documents included in this study may not fully represent the entirety of the literature on AI and educational leadership. This limitation is exacerbated by covering only one database (Scopus) with some specific keywords. Therefore, future research could complement this study by covering more databases, non-English studies, and using more search keywords.

Implications

Despite these limitations, we remain optimistic that the results of our study can serve as inspiration for future research synthesis. We encourage further investigations into the intersection of leadership and AI, paying particular attention to the usage of terms and concepts and their interconnectedness. The current study makes significant contributions (theoretical and practical) to the literature in the field of AI and educational leadership, offering three key insights.

From a theoretical perspective, particularly from a temporal and contextual factor of a theory (Dubin, 1978; Whetten,1989), this present study (based on Figures 2 and 3) can contribute to developing theories around the factors that might catalyze AI in educational leadership research, as well as cross-country collaboration to promote the safe and effective of AI in educational leadership. For instance, as the discussion of AI ethics varies from a country to another, economic theory can be used to discuss the regulatory frameworks and laws of integrating AI in educational leadership. Particularly, the study highlights a substantial imbalance in knowledge coverage, particularly regarding the global south and non-Western societies. This observation underscores the need for future research to address misrepresented contexts and foster intercultural collaboration that transcends spatial and cultural boundaries. From a conceptual landscape, the present study (based on Figures 6 and 7) provided the landscape of research field on AI in educational leadership, and where it is headed. This reveals several dimensions that should be considered when developing a theoretical framework of AI in educational leadership.

From a practical perspective, the study argues, based on a thoughtful analysis and ontological understanding (see Figures 5 and 6), that AI, if implemented carefully, can enhance decision making and leadership aligned with a modern understanding of the learning process and goals. It emphasizes that AI technologies alone will not automatically enhance the educational system; deliberate efforts are essential to transform education into a humanistic twenty-first-century enterprise. The role of educational leadership is deemed crucial in this transformation, positioning technology as a means to redefine educational goals within a humanistic framework, thereby redirecting technology to create a twenty-first-century humanistic and effective educational enterprise. Besides, the present study revealed (based on Table 1) the needs for more efforts to achieve some particular SDGs (e.g., SDGs 5 and 10) when implementing AI in educational leadership, hence achieving the goals of United Nations (UN) by 2030.

In conclusion, the paper advocates for new thinking in response to a changing world, emphasizing the dual and multiplicative responsibility imposed on the social system entrusted with the distribution and sharing of public knowledge in a knowledge-rich society. This underscores the need for reframing and reaffirming the school leadership role in the current era, aligning with the demands of a new world. In practical terms, building on this analysis, we propose a redefinition of the fundamental problems within the education system, aiming to leverage technology for their resolution. This goes beyond mere improvement; it calls for a transformative process requiring a sustained effort to change the behavior of the entire educational community. Educational leadership emerges as a pivotal force in this transformation, tasked with leading the way in redefining the very nature and goals of the educational enterprise. The utilization of knowledge technologies becomes instrumental in reshaping education into a twenty-first-century humanistic enterprise.

As highlighted, in the face of demanding innovations like innovative learning tools and knowledge technologies that transcend the traditional constraints of place and time, leaders are likely to adapt to systemic changes. This necessitates the display of an open mindset, reflecting a new form of leadership known as facilitative leadership. This leadership style encourages participation, enables continuous operation, recognizes diverse values, unleashes the unlimited potential of learning, and mobilizes collaborative efforts both within and outside the educational entity. The synergy between knowledge technologies and ubiquitous communication networks creates fertile ground for creativity and collaboration. This further raises a fundamental question: *How do leaders envision education in 2051 considering the ubiquitous nature of AI in the educational arena?*

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